

IFY STATISTICS | SETS, NUMBERS, COUNTING AND ARITHMETIC

01 SETS AND NOTATION

Set: a well-defined collection of distinct objects. Membership is clear; repeats count once: $\{2, 2, 4\} = \{2, 4\}$.

Braces { } enclose a set. An **element/member** is one object in it.

$\in / \notin$ Is an element of / is not an element of.

$U / \emptyset$ **Universal set:** everything considered. **Empty set:** no elements; $\emptyset \neq \{0\}$.

$n(A)$ **Cardinality:** number of elements. $A = \{2, 4, 6\}$ gives $n(A) = 3$.

$A \cap B$ **Intersection:** in both sets.

$A \cup B$ **Union:** in either set or both.

A^c / A' **Complement:** in U but not A .

$A \subseteq B$ **Subset:** every element of A is in B . $A \subset B$: **proper subset**; B has more.

Venn diagram: circles show sets; the **overlap** is the intersection. **Only A:** in A , not B . **Neither:** outside both but inside U .

02 NUMBER SYSTEMS

Natural Counting numbers: 1, 2, 3, ...

Whole Zero and counting numbers: 0, 1, 2, ...

Integer No fractional part: -5, 0, 12.

Rational p/q with integer p , q and $q \neq 0$. $-2 = -2/1$; $0.25 = 1/4$.

Irrational Cannot be written as p/q ; decimal never ends or repeats: $\sqrt{2}$, π .

Real Any rational or irrational number.

Terminating decimal: ends (0.25).

Recurring/repeating: repeats (0.272727...).

Fraction: numerator (top) $\div$ denominator (bottom; not zero). **Equivalent forms:** $3/4 = 0.75 = 75\%$.

Classify: simplify first, then name the smallest number family: $\sqrt{9} = 3 \rightarrow$ natural.

Symbols: = equal; $\neq$ not equal; $\approx$ approximately. $>$ greater than; $<$ less than; $\rightarrow$ next step.

03 COUNTING METHODS

Outcome: one complete result. **Stage/slot:** one choice to make. In a **tree diagram**, branches show choices; a complete path gives an outcome.

Basic Counting Principle: multiply the valid choices at each stage. 3 shirts $\times$ 2 trousers = **6 outfits**.

Restriction: a rule limiting choices. A **leading digit** cannot be 0 in a multi-digit number; a PIN may start with 0.

Repeats allowed / with replacement: reuse choices. **No repeats / without replacement:** used items become unavailable.

Factorial: $n! = n \times (n-1) \times \dots \times 1$
 $5! = 120$; $0! = 1$.

PERMUTATION (nPr): ORDER MATTERS

$$P(n, r) = \frac{n!}{(n-r)!}$$

Different roles or order create different outcomes. $P(5, 3) = 5 \times 4 \times 3 = \mathbf{60}$.

COMBINATION (nCr): GROUP MATTERS

$$C(n, r) = \frac{n!}{r!(n-r)!}$$

Changing the order does not create a new group. $C(5, 3) = \mathbf{10}$.

n = distinct items available; **r** = items selected. These formulas assume **no repeats**. $(n-r)!$ cancels unused factors; $r!$ removes repeated orders.

04 ARITHMETIC AND ORDER

+ addition / sum / total **-** subtraction / difference
 $\times$ multiplication / product **$\div$** division / quotient
"Of" often means $\times$; "per" often means $\div$.

Factor: divides exactly; **common** = shared.

GCD/HCF: greatest common divisor / highest common factor.

Power: in 5^2 , base = 5; exponent = 2.

Squared/cubed: powers $2/3$. A **root** reverses a power: $\sqrt{9} = 3$.

PEMDAS / BODMAS

Brackets / parentheses $\rightarrow$ exponents / orders
Then $\times / \div$, then $+$ / $-$; each pair **left to right**.
 $24 \div 6 \times 2 = \mathbf{8}$. **Evaluate:** calculate the value.

05 ALGEBRA AND EQUATIONS

Algebra uses letters to represent numbers.

$$-2x + 5$$

x: variable **-2:** signed coefficient

5: constant **-2x and +5:** terms

Like terms have the same variables and powers.

Expression: no equals sign. **Equation:** expressions joined by =. **LHS/RHS:** left/right-hand side.

Simplify: combine like terms: $5x - 2x = 3x$.

Substitute: replace a variable with its value.

Expand/distribute: multiply each bracket term:
 $2(y - 1) = 2y - 2$.

Inverse operations undo each other: $+/-$, $\times/\div$.

Balance principle: do the same to both sides.

Isolate: leave the variable alone. **Solve:** find a value that makes the equation true. **Check/verify:** substitute the solution back.

06 QUADRATIC EQUATIONS

Linear / quadratic / cubic: highest variable power is 1 / 2 / 3.

Standard form:

$$ax^2 + bx + c = 0 \quad (a \neq 0)$$

a multiplies x^2 ; **b** multiplies x ; **c** is constant. Keep every sign when finding a, b and c.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$\pm$: calculate once with + and once with -.

The whole numerator is divided by **2a**.

Discriminant: $b^2 - 4ac$. Positive $\rightarrow$ two real roots; zero $\rightarrow$ one repeated root; negative $\rightarrow$ no real roots. A **root** is a solution.

Factorise: rewrite as multiplied factors.

Zero-product rule: $AB = 0 \rightarrow A = 0$ or $B = 0$.

Before calculating: check the signed coefficients and brackets. Real-number roots lie on the number line.

07 UNITS AND CONVERSIONS

Quantity/measurement: amount and property.

Unit: measurement standard. Length: mm, cm, m, km, in, ft; mass: g, kg, oz, lb; volume: mL, L, fl oz.

Convert: express the same measurement in another unit. A **conversion factor** is a ratio of equivalent quantities, equal to 1.

$$750 \text{ mL} \times \frac{1 \text{ L}}{1000 \text{ mL}} = 0.75 \text{ L}$$

Unit cancellation: matching units above/below cancel. **Decimal places (d.p.):** digits after the decimal point. **Capacity:** maximum allowed.

08 PERCENTAGES

Percentage: out of 100. **Part = rate $\times$ whole**, with the rate written as a decimal.
 $25\% \text{ of } 80 = 0.25 \times 80 = 20$.

Starting / new: before / after change.

Amount of change: the difference.

$$\text{INCREASE} \quad \frac{\text{New} - \text{starting}}{\text{Starting}} \times 100\%$$

$$\text{DECREASE} \quad \frac{\text{Starting} - \text{new}}{\text{Starting}} \times 100\%$$

The denominator is the **starting value**; it cannot be 0.

Discount: amount removed; **sale price** remains. **Tax:** amount added to get the total price.

09 RATIOS AND PROPORTIONS

Ratio: comparison in a stated order. A **colon :** means "to". **Equivalent ratios:** the same comparison; $3:4 = 21:28$.

Scale factor: multiply/divide each part equally.

Simplify: use a common factor; $45:60 = 3:4$.

Total parts: add the ratio numbers. Divide 70 in **2:5**:
 7 parts ; $70 \div 7 = 10$ per part $\rightarrow$ **20 and 50**.

Proportion: equal ratios. **Cross-multiply:**
 $3/4 = 21/g \rightarrow 3g = 4 \times 21 \rightarrow g = 28$.

Part-to-part: one group / another.

Part-to-whole: one group / combined total.

Ratio $4:5$ gives **4/5** versus **4/9**.

"Out of" compares a part with the whole.